

Target Trial Emulation (TTE) vs Just Analyzing the Data

The Two Approaches: A Tale of Two Analysts

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Imagine we have a large electronic health record (EHR) dataset and we want to answer the question: “Does taking statins prevent cardiovascular disease (CVD)?”

Approach 1: “Just Analyzing the Data”

A traditional approach might look like this:

1. **Data First:** Open the dataset. Identify a variable for “statin use” (e.g., `statin_ever_used`) and an outcome variable for “CVD.”
2. **Model:** Choose a regression model. Let’s use logistic regression: $\text{CVD} \sim \text{statin_ever_used} + \text{age} + \text{sex} + \text{diabetes}$.
3. **Result:** Fit the model. Look at the odds ratio for `statin_ever_used`. If it’s less than 1 and the p-value is significant, conclude that statins are protective.

This seems straightforward, but it is filled with hidden pitfalls. The causal question is vague, and the analysis is susceptible to numerous biases that are difficult to see without a more structured framework.

Approach 2: Target Trial Emulation (TTE)

The TTE framework forces us to change our entire workflow.

- **Mindset Shift:** We **start with the question, not the data**. Our first step is to design the ideal, hypothetical randomized controlled trial (RCT) we *wish* we could run to answer our question. This is our “target trial.” Only after we have a complete “protocol” for this ideal trial do we turn to the observational data and see how well we can **emulate** it.

This disciplined process makes our assumptions explicit and helps us proactively design an analysis that avoids common biases.

Tutorial: Building the Target Trial Protocol

The protocol of any trial (real or hypothetical) has key components. Let's define them for our statin/CVD question and contrast the TTE approach with the "just analyze" approach at each step.

1. Eligibility Criteria

- **The Question:** Who would be eligible to enroll in our ideal trial?
- **"Just Analyze" Approach:** This is often an afterthought. We might just use all patients in the dataset who don't have missing values for our variables of interest.
- **TTE Approach:** We must be explicit. For our statin trial, we might specify:
 - *Inclusion:* Patients aged 40-75.
 - *Exclusion:* Patients with a history of CVD at baseline (since we're studying prevention). Patients without continuous health plan enrollment for the prior year (to ensure we have their baseline data).
 - **Why it matters:** Excluding prevalent users (those already on statins) and requiring a clean baseline period is crucial for avoiding selection bias and ensuring we are comparing apples to apples.

2. Treatment Strategies

- **The Question:** What are the precise interventions we are comparing?
- **"Just Analyze" Approach:** We might use a vague `statin_ever_used` variable. But what does this mean? Did they take one pill? Take it for 10 years? This ambiguity is a major source of error.
- **TTE Approach:** We must specify the strategies precisely, as in a real trial.
 - **Strategy 1:** Initiate a statin (e.g., atorvastatin, simvastatin) at baseline.
 - **Strategy 2:** Do not initiate a statin at baseline.
 - **Why it matters:** This pushes us towards a "new-user" design, where we only include patients at the moment they become eligible to start the therapy. This is a powerful tool to prevent biases.

3. Assignment Procedure

- **The Question:** How are individuals assigned to a treatment strategy?
- **“Just Analyze” Approach:** The assignment is whatever happened in the real world. We hope to “control for confounding” by adding covariates to a regression model.
- **TTE Approach:** In our ideal target trial, we would assign treatment **randomly** at baseline. In the real world, this is obviously not the case; doctors assign statins for reasons (confounding). Our task is to **emulate randomization**. We do this by adjusting for a sufficient set of baseline confounders (identified by our DAG!) using methods like:
 - **Inverse Probability of Treatment Weighting (IPTW):** Creating a “pseudo-population” where the confounders are no longer associated with treatment assignment.
 - **Matching:** Matching treated individuals to untreated individuals with similar baseline characteristics.
 - **Why it matters:** This makes the goal of “controlling for confounding” a deliberate design choice, not just a modeling exercise. We are explicitly trying to create groups that are exchangeable, just like in a real RCT.

4. Follow-up Period

- **The Question:** When does follow-up begin and end?
- **“Just Analyze” Approach:** The start of follow-up is often poorly defined. Does it start when the patient enters the database? Or when they first use a statin?
- **TTE Approach:** The start of follow-up (**time zero**) must be the same for everyone: the moment of treatment assignment (or emulation of assignment).
 - For new users, this is the date of their first statin prescription.
 - For non-users, we can assign a similar “index date.”
 - Follow-up ends at a pre-specified time (e.g., 5 years), or when CVD occurs, or when they leave the health plan (censoring).
 - **Why it matters:** Properly defining time zero is the **single most effective way to prevent immortal time bias**. This bias occurs when a period of follow-up is misclassified as “unexposed” when it should be “not yet exposed,” leading to a spurious protective effect.

5. Outcome

- **The Question:** How do we define the outcome?
- **TTE Approach:** We need to specify the outcome just as we would in a trial protocol (e.g., “first diagnosis of a myocardial infarction or stroke, identified via ICD-10 codes during an inpatient hospital stay”). This clarity reduces measurement error.

6. Causal Contrasts (Estimand)

- **The Question:** What is the precise causal effect we are trying to estimate?
- **“Just Analyze” Approach:** This is often implicit in the model output (e.g., an odds ratio).
- **TTE Approach:** We specify our estimand up front. The two main choices are:
 - **Intention-to-Treat (ITT) effect:** The effect of *assigning* the treatment strategy at baseline, regardless of whether the patient adheres to it later. This is often emulated in observational data.
 - **Per-Protocol effect:** The effect of *adhering* to the treatment strategy over time. This is more complex and requires advanced methods like Marginal Structural Models (MSMs) to handle time-varying confounding.
 - **Why it matters:** Being explicit about the estimand dictates the correct analysis plan and prevents misinterpretation of the results.

7. Analysis Plan

- **The Question:** How will we estimate the causal contrast?
- **“Just Analyze” Approach:** Fit a single regression model.
- **TTE Approach:** The analysis plan flows directly from all the previous design choices. If emulating an ITT effect, we might use a weighted Cox model (using IPTW) that compares the two groups from time zero. If estimating a per-protocol effect, we would need to specify how we will handle treatment switching and censoring.

Summary: Why Target Trial Emulation is Different

Feature	“Just Analyzing Data”	Target Trial Emulation
Starting Point	The dataset and its variables	A precise causal question & an ideal trial protocol
Assumptions	Often implicit, hidden in the model	Explicitly stated in the protocol design
Bias Avoidance	Reactive (hope regression “controls” for it)	Proactive (design choices to prevent bias)
Transparency	Low; it’s hard to know exactly what was compared	High; the emulated protocol is the roadmap for the analysis
Rigor	Variable; prone to hidden errors	High; provides a systematic framework

In short, target trial emulation is not a statistical method. It's a **structured way of thinking**. It's the overarching framework that guides you to ask the right questions, design a sound observational analysis, choose the right statistical methods, and avoid preventable biases. It turns a simple data analysis into a credible scientific investigation.

Trial Emulation Example

Here are the seven components of a target trial designed to answer the question: **"What is the effect of obtaining a bachelor's degree on personal income?"**

1. Eligibility Criteria

This defines the population for our hypothetical trial. We can't include everyone. We need a specific cohort of people who are at the point of making the decision.

- **Eligible Population:** All individuals who graduated from a U.S. high school in a specific year (e.g., 2010), are between 17-19 years old at graduation, and have not yet enrolled in a 4-year college program.
- **Exclusions:** Individuals who are not eligible to apply to a 4-year college or who already possess an associate's degree or other advanced vocational certification.

2. Treatment Strategies

These are the precise interventions we want to compare. "Getting a degree" is too vague.

- **Strategy 1 (Treated):** Enroll in a 4-year bachelor's degree program within one year of high school graduation and successfully obtain a bachelor's degree within six years of high school graduation.
- **Strategy 2 (Control):** Do not obtain a bachelor's degree within six years of high school graduation. (This group may enter the workforce, attend community college, or pursue vocational training.)

3. Assignment Procedure

In an ideal world, we would randomize. In our emulation, we mimic that randomization.

- **Ideal Trial:** Eligible individuals would be randomly assigned at high school graduation to either pursue the "degree" strategy or the "no degree" strategy.

- **Emulation:** We use observational data and adjust for the baseline factors that influence both the decision to attend college and future income. This is our **confounder set**. We would use methods like propensity score weighting or matching to balance the two groups on key baseline variables measured at high school graduation, such as:
 - High School GPA and class rank
 - Standardized test scores (SAT/ACT)
 - Family income and socioeconomic status
 - Parental education level
 - High school quality and resources

4. Follow-up Period

This defines the timeline for our study, and **time zero** is critical.

- **Start of Follow-up (Time Zero):** The clock starts for everyone at the same moment: right after high school graduation (e.g., September 1, 2010). This is the point of (emulated) randomization.
- **End of Follow-up:** The follow-up period ends when the outcome is measured. A reasonable point would be 12 years after high school graduation, when individuals are approximately 30 years old and have had time to establish themselves in their careers.

5. Outcome

This is the specific event we are measuring.

- **Primary Outcome:** Annual personal pre-tax income at age 30, measured in U.S. dollars. The data source (e.g., tax records, survey data) must be specified.

6. Causal Contrasts (Estimand)

This is the specific statistical quantity we aim to estimate.

- **Estimand:** The **Average Treatment Effect (ATE)**. Specifically, we want to calculate the difference in mean annual income at age 30 between the group assigned to Strategy 1 and the group assigned to Strategy 2. This is an “intention-to-treat” estimand, as it measures the effect of being assigned to a strategy at baseline, regardless of eventual adherence (e.g., some people assigned to the college track might drop out).

7. Analysis Plan

This outlines the statistical steps to estimate the causal contrast.

- **The Plan:**

1. Among the eligible cohort of 2010 high school graduates, fit a model (e.g., logistic regression) to estimate each person's probability of attending college based on the baseline confounders (GPA, family income, etc.). This yields the propensity score.
2. Use these propensity scores to calculate inverse probability of treatment weights (IPTW) to create a pseudo-population where the baseline confounders are balanced between the two groups.
3. In this weighted pseudo-population, calculate the mean income at age 30 in the group that obtained a degree and the group that did not.
4. The difference between these two means is the estimate of the causal effect. We would then calculate a 95% confidence interval around this estimate.